In [1]:
```
%run Latex_macros.ipynb
```

$$ \newcommand{\x}{\mathbf{x}} \newcommand{\tx}{\tilde{\x}} \newcommand{\y}{\mathbf{y}} \newcommand{\b}{\mathbf{b}} \newcommand{\c}{\mathbf{c}} \newcommand{\e}{\mathbf{e}} \newcommand{\z}{\mathbf{z}} \newcommand{\h}{\mathbf{h}} \newcommand{\u}{\mathbf{u}} \newcommand{\v}{\mathbf{v}} \newcommand{\w}{\mathbf{w}} \newcommand{\V}{\mathbf{V}} \newcommand{\W}{\mathbf{W}} \newcommand{\X}{\mathbf{X}} \newcommand{\KL}{\mathbf{KL}} \newcommand{\E}{{\mathbb{E}}} \newcommand{\Reals}{{\mathbb{R}}} \newcommand{\ip}{\mathbf{{(i)}}} % % Test set \newcommand{\xt}{\underline{\x}} \newcommand{\yt}{\underline{\y}} \newcommand{\Xt}{\underline{\X}} \newcommand{\perfm}{\mathcal{P}} % % \ll indexes a layer; we can change the actual letter \newcommand{\ll}{l} \newcommand{\llp}{{(\ll)}} % \newcommand{Thetam}{\Theta_{-0}} % CNN \newcommand{\kernel}{\mathbf{k}} \newcommand{\dim}{d} \newcommand{\idxspatial}{{\text{idx}}} \newcommand{\summaxact}{\text{max}} \newcommand{idxb}{\mathbf{i}} % % % RNN % \tt indexes a time step \newcommand{\tt}{t} \newcommand{\tp}{{(\tt)}} % % % LSTM \newcommand{\g}{\mathbf{g}} \newcommand{\remember}{\mathbf{remember}} \newcommand{\save}{\mathbf{save}} \newcommand{\focus}{\mathbf{focus}} % % % NLP \newcommand{\Vocab}{\mathbf{V}} \newcommand{\v}{\mathbf{v}} \newcommand{\offset}{o} \newcommand{\o}{o} \newcommand{\Emb}{\mathbf{E}} % % \newcommand{\loss}{\mathcal{L}} \newcommand{\cost}{\mathcal{L}} % % \newcommand{\pdata}{p_\text{data}} \newcommand{\pmodel}{p_\text{model}} % % SVM \newcommand{\margin}{{\mathbb{m}}} \newcommand{\lmk}{\boldsymbol{\ell}} % % % LLM Reasoning \newcommand{\rat}{\mathbf{r}} \newcommand{\model}{\mathcal{M}} \newcommand{\bthink}{\text{}} \newcommand{\ethink}{\text{}} % % % Functions with arguments \def\xsy#1#2{#1^#2} \def\rand#1{\tilde{#1}} \def\randx{\rand{\x}} \def\randy{\rand{\y}} \def\trans#1{\dot{#1}}

\def\transx{\trans{\x}} \def\transy{\trans{\y}} % \def\argmax#1{\underset{#1}{\operatorname{argmax}}} } \def\argmin#1{\underset{#1}{\operatorname{argmin}}} } \def\max#1{\underset{#1}{\operatorname{max}}} } \def\min#1{\underset{#1}{\operatorname{min}}} } % \def\pr#1{\mathcal{p}(#1)} \def\prc#1#2{\mathcal{p}(#1 \; | \; #2)} \def\cnt#1{\mathcal{count}_{#1}} \def\node#1{\mathbb{#1}} % \def\loc#1{{\text{##}{#1}}} % \def\OrderOf#1{\mathcal{O}\left({#1} \right)} % % Expectation operator \def\Exp#1{\underset{#1}{\operatorname{\mathbb{E}}} } % % VAE \def\prs#1#2{\mathcal{p}_{#2}(#1)} \def\qr#1{\mathcal{q}(#1)} \def\qrs#1#2{\mathcal{q}_{#2}(#1)} % % Reinforcement learning \newcommand{\Actions}{{\mathcal{A}}} \newcommand{\action}{a} \newcommand{\act}{a} \newcommand{\States}{{\mathcal{S}}} \newcommand{\stateseq}{S} \newcommand{\state}{s} \newcommand{\Rewards}{{\mathcal{R}}} \newcommand{\rewseq}{R} \newcommand{\rew}{r} \newcommand{\transp}{P} \newcommand{\statevalfun}{v} \newcommand{\actvalfun}{q} \newcommand{\disc}{\gamma} \newcommand{\advseq}{\mathbb{A}} % % \newcommand{\floor}[1]{\left\lfloor #1 \right\rfloor} \newcommand{\ceil}[1]{\left\lceil #1 \right\rceil} % % $$

Case study: Post-training a model to reason

A *reasoning model* provides responses with a distinctive style

- format
- process
  - long Chain-of-Thought (CoT): step-by-step reasoning
  - *reflection*: looking back at the response so far, and evaluating the solution and strategy
  - *revision*: adapting/changing the current response and strategy

Reasoning Format example

Instruction:
"Explain step-by-step how to find the greatest common divisor (GCD) of 48 and 18."

Expected Reasoning Format:

1. **State the problem clearly:** "We want to find the GCD of 48 and 18."
2. **Describe the method or approach:** "We will use the Euclidean algorithm."
3. **Stepwise execution:**
    - Step 1: Divide 48 by 18, the remainder is 12.
    - Step 2: Divide 18 by 12, the remainder is 6.
    - Step 3: Divide 12 by 6, the remainder is 0.
4. **Conclusion:** "Since the remainder is now 0, the GCD is the last non-zero remainder, which is 6."

Formatted Output:
"We want to find the GCD of 48 and 18. Using the Euclidean algorithm,
Step 1: 48 divided by 18 leaves a remainder of 12.
Step 2: 18 divided by 12 leaves a remainder of 6.
Step 3: 12 divided by 6 leaves a remainder of 0.
Therefore, the GCD of 48 and 18 is 6."

-

Reasoning behavior is something that is instilled in post-training

- Not the natural behavior of an LLM or Assistant

We will demonstrate how this is done.

We will use Reinforcement Fine Tuning (RFT).

As you may have noticed in our previous section, RFT has at least two steps

- Supervised Fine Tuning
- Reinforcement Learning
    - usually with Preference Data

We will review each step and explain why they are both necessary.

Supervised Fine Tuning + Reinforcement Learning: Why both ?

It may seem confusing to need both SFT and RL.

Very loosely

- SFT is used to teach the base model the *style* of a reasoning response
    - syntax
    - surface level
- RL is used to ensure that the reasoning response behaves according to *valid logic*
    - semantic
    - deeper level

Supervised Fine Tuning is more about

- *imitating* training examples
    - can *overfit* to training examples
        - it is the nature of the Loss function
            - bounded below by 0
- than *understanding* a process
    - fail to generalize beyond training examples

Reinforcement Learning creates a deeper understanding

- iterative feedback via rewards
    - maximizing return: can always try to improve
    - no clear upper bound

On a more practical level, to instill reasoning behavior

- SFT
    - requires *many* training examples
    - typically: human labeled
- RL
    - needs fewer examples
    - iterative improvement with each reward
    - can *re-use* the same example to improve further
        - reward can increase with each re-use

Supervised Fine Tuning: avoiding the ”cold start” problem

It would seem that RL is superior to SFT

- why is SFT necessary ?

A partial answer is that

- reasoning responses
- are *very different* than the response to the same prompt on a base model
    - it is "out of distribution"

Reinforcement Learning struggles with the "out of distribution" responses of the training examples

- Sparse rewards
    - trajectory reward, no intermediate reward

SFT is very good at adapting the base model's outputs to the "new distribution"

- the different style of a reasoning response

Hence SFT is usually used as an initial step.

This overcomes the *Cold Start* problem.

Interestingly, SFT instills

- the *format*
    - step by step
- and *patterns*
    - reflection, revision
- *not necessarily* correctness of reasoning !
    - or at least: correct w.r.t. training examples
    - poor generalization

SFT creates a stable base upon which RL can learn to generalize.

Stage	Purpose in Reasoning Induction	Training Signal/Data	Strengths	Limitations
SFT	Learn reasoning formats and step-by-step logic	Paired (instruction, reasoning chain) examples	Provides stable, structured output	Limited generalization, mimicry
RL (e.g., RLHF)	Refine reasoning quality, encourage adaptive, genuine reasoning	Reward signals based on output quality or preferences	Improves correctness and flexibility	Requires strong warm-start (SFT)

References for SFT and RL stages

- SFT or RL? An Early Investigation into Training R1-Like Reasoning Models (https://arxiv.org/html/2504.11468v1)
- Dissecting Mathematical Reasoning for LLMs Under Reinforcement Learning (https://arxiv.org/html/2506.04723v1)
- Beyond Next-Token Prediction: How Post-Training Teaches LLMs to Reason (https://toloka.ai/blog/how-post-training-teaches-llms-to-reason/)

DeepSeek: investigating the Cold Start problem

DeepSeek-R1 is a well known reasoning model.

Its development included experiments centered around the necessity of the SFT step.

Specifically

- the authors tried an *RL only* (no SFT) approach
- resulting in a reasoning model DeepSeek-R1-Zero
    - strong reasoning
    - inconsistent formatting
        - mixed English/Chinese output !

This confirmed the need for

- at least a *small* number of training examples for SFT
- to overcome the Cold Start

But the inconsistent DeepSeek-R1-Zero was still very useful

- was prompted to create reasoning responses
- these inconsistent reasoning
- were filtered/curated by the human developers to overcome format issues
- in order to create an *expanded set* of SFT examples !

This bootstrapping resulted in a large SFT training set

- 600K examples
- which improved the SFT step greatly
    - adherence to format
    - instruction-following
- but the SFT-only (i.e., without the subsequent RL step) model
    - still failed to
        - reason correctly
        - generalize out of sample

This validate the necessity of the RL step.

Synthetic generation of reasoning examples

Using this idea, we can fine-tune (via SFT) a base model

- to produce responses in reasoning **format**
- not necessarily logically correct
- not necessarily using the right process

This fine-tuned model becomes

- an abundant source of training examples
- for the SFT step of RFT

Note

Distinguish between

- the SFT-model trained to produce examples for RFT
- the base model that uses these examples for the initial SFT step of RFT

Here is a hypothetical one-shot prompt

- to create a new example of a question
- and a reasoning response

Its goal is to tune the model to

- produce responses
- in the desired format
    - structured sections for major steps
    - step by step answer strategy

One-shot prompt

```
Below is an example showing how to answer a question with clear structured reas
oning including labeled sections.

For each new question you invent, provide the reasoning answer in the same labe
led format.

**Instruction:**
"Explain step-by-step how to find the greatest common divisor (GCD) of 48 and 1
8."

**Expected Reasoning Format:**
1. **State the problem clearly:** "We want to find the GCD of 48 and 18."
2. **Describe the method or approach:** "We will use the Euclidean algorithm."
3. **Stepwise execution:**
   - Step 1: Divide 48 by 18, the remainder is 12.
   - Step 2: Divide 18 by 12, the remainder is 6.
   - Step 3: Divide 12 by 6, the remainder is 0.
4. **Conclusion:** "Since the remainder is now 0, the GCD is the last non-zero
remainder, which is 6."
```

The above is another example

- of Synthetic Data generation in action
- use In-Context Learning

Reference to DeepSeek-R1

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning from Diverse Feedback (https://arxiv.org/abs/2501.12948)

Example: RFT = SFT + RL with Preference Data

To make all the steps clear, we provide an example of each step (and sub-step in some cases).

These examples reflect how

- SFT focuses on getting the model to produce structured, formatted reasoning
    - by learning from labeled examples
- RL uses reward feedback
    - to push the model
    - toward more accurate, meaningful, and high-quality reasoning

Example: SFT Training Example (Instruction + Detailed Reasoning)

Here is an input example used in the SFT step

- to train the base model to produce the response in **correct format**

Input (Instruction + Question):
"Explain step-by-step how to solve the equation 2x + 3 = 9."

Output (Reasoning Steps):
"Step 1: Subtract 3 from both sides: 2x + 3 - 3 = 9 - 3, which simplifies to 2x = 6.
Step 2: Divide both sides by 2: 2x / 2 = 6 / 2, so x = 3."

Note: This example teaches the **format and structure** of reasoning—how to break a problem down into clear steps.

-

Note that the SFT can produce outputs

- in the correct format
- but with flawed logic
    - RL instills correct logical reasoning

For example:

Input:
"Explain why the Earth revolves around the Sun."

Output:
"The Earth moves around the Sun because the Sun is bigger and pulls the Earth with its big gravity."

Note: The format is a coherent explanation, but the reasoning may be oversimplified or imprecise.

Example: RL Training Data Example (Preferences/Rewards)

Here is an example of the input to **train the reward/preference model**

The trained model can then be used to create a Preference Data set for the RL step.

Candidate Outputs for the same input:

- **Output A:**
  "Step 1: Subtract 3 from both sides: 2x = 6. Step 2: Divide both sides by 2: x = 3."
  (Concise and logically correct.)
- **Output B:**
  "Subtract 3 from both sides and divide by 2, so x = 3. This is because math."
  (Vague and incomplete reasoning.)

Reward Signal:
Output A is *preferred* and given a higher reward; Output B is penalized for lack of detailed and correct reasoning.

Example: RL Encouraging Improved Reasoning Quality

Here is an example of the input to the RL step

- an example of Preference Data
- where the reward for the preferred responses is higher than for the non-preferred response

Candidate Outputs:

- **Output A:**
  "The Earth revolves around the Sun due to the gravitational force described by Newton's law of universal gravitation, where the Sun's mass exerts a force on Earth keeping it in orbit."
- **Output B:**
  "The Sun is big and bright, so Earth moves around it."

Reward:
Output A receives higher reward for scientifically accurate and logically sound reasoning, refining the correctness beyond SFT's imitation.

Comparison: SFT, RL, RFT

SFT and RL are different methods for fine tuning an LLM.

- RFT combines an initial SFT with a subsequent RL

The distinction becomes every blurrier

- when RL has intermediate rewards
- rather than a single trajectory reward

It is sometimes possible to cast a task into a form appropriate for either SFT or RL with per-step rewards

- Next token prediction
    - SFT: Cross Entropy Loss for every step
    - RL: Per-step reward
        - +1 reward for correct prediction/0 for incorrect prediction

But the choice of which method to use is often dependent on

- the task
- the available training data

It is hard to be precise, but here are some thematic comparisons.

SFT

- encourages imitation of the label of an example
    - exact match
- enforces formatting/structure of response
- "surface" level correctness
- well-suited to precisely-defined tasks
    - with *objective* measures of success
    - quantitative measure

RL

- allows multiple "correct" answers
    - which may be ranked
- "deeper" understanding/generalization
- well-suited to more loosely-defined tasks
    - with *subjective* measures of success
    - *qualitative* measures

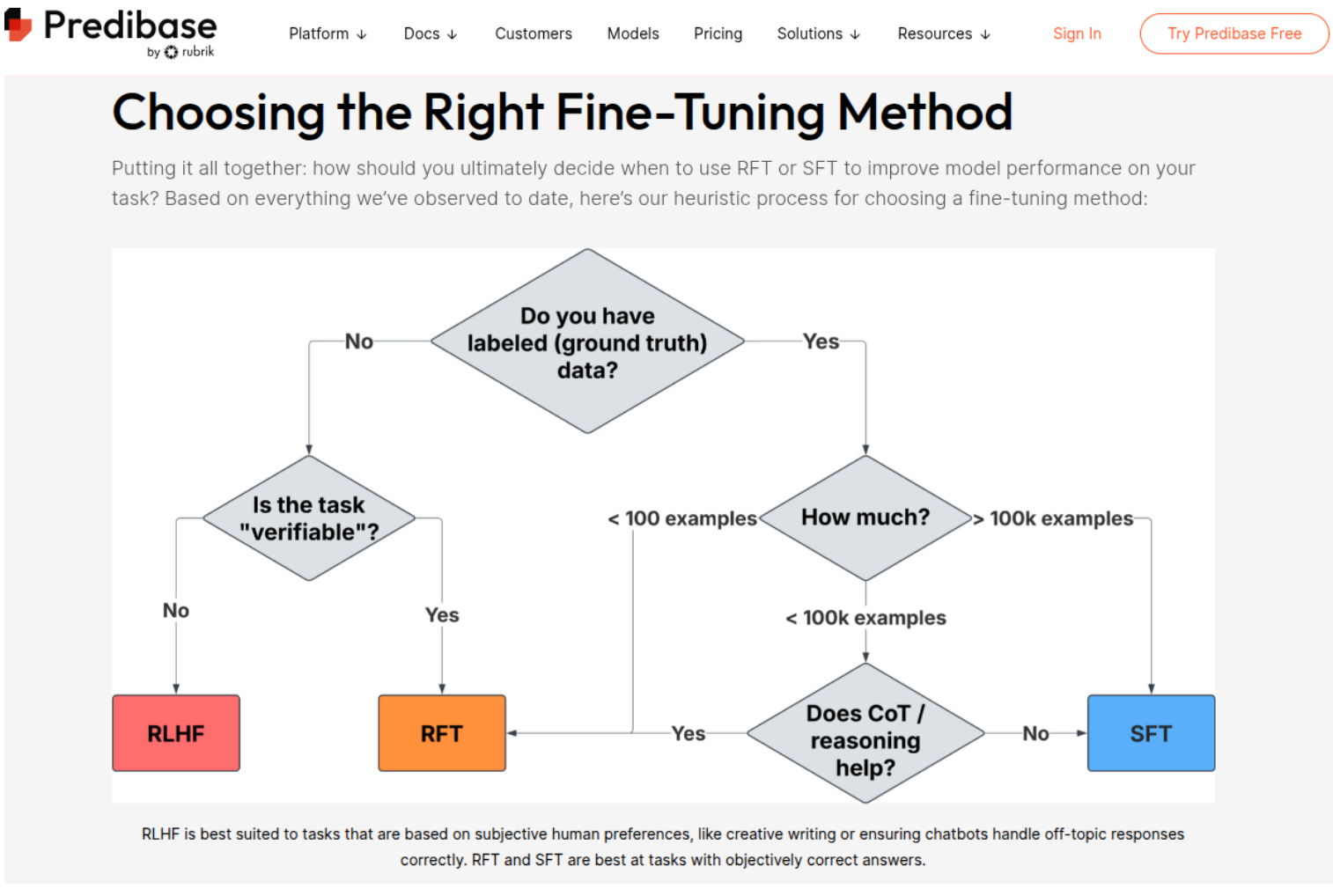
In terms of training data

- SFT imitation requires *lots* of training examples
    - exploration of alternatives doesn't come into play
- RL can often be accomplished in a very small number of training examples
    - exploration encourage

SFT and RL are *complementary* methods for fine tuning an LLM.

Criteria	SFT	RFT/RLHF
Task type	Objective, well-defined, clear correct answer tasks	Subjective, ambiguous, or value-laden tasks
Data availability	Large, high-quality labeled datasets available	Little/no labeled data, but feedback/preference signals are available
Training complexity	Simpler (labeled pairs)	More complex (reward model, RL optimization)
Desired outcome	Accuracy, task performance, factual correctness	Human preference alignment, style, quality, safety
Overfitting risk	Higher, if data is limited	Lower; learns general behavior from rewards
Generalization	Prone to memorization	Promotes adaptability, nuanced behaviors
Cost/resource needs	Lower; less human-in-the-loop need	Higher; human feedback collection and more computation
Ideal use cases	Translation, classification, summarization, retrieval	Chatbots, open-domain QA, content moderation, dialog

Here is one rubric:



Reference: [https://predibase.com/blog/how-reinforcement-learning-beats-supervised-fine-tuning-when-data-is-scarce (https://predibase.com/blog/how-reinforcement-learning-beats-supervised-fine-tuning-when-data-is-scarce)](https://predibase.com/blog/how-reinforcement-learning-beats-supervised-fine-tuning-when-data-is-scarce)

Use of SFT as first phase of RFT

The preliminary SFT phase of RFT serves several purposes

- move the LLM's distribution to the *format* of tasks for RL to learn
    - primes the RL phase with correctly formated examples
- creates examples with different rewards
    - RL learns from the *contrasts* between high and low rewards
- "bootstrap" the RL training dataset
    - uses iterative SFT
        - uses a weaker SFT-tuned model
        - to create synthetic training examples for a stronger model

References for RFT vs SFT

- Why Reinforcement Learning Beats SFT with Limited Data - Predibas (https://predibase.com/blog/how-reinforcement-learning-beats-supe tuning-when-data-is-scarce)
- Preference Alignment vs Supervised Fine-Tuning in LLM Training (https://www.rohan-paul.com/p/preference-alignment-vs-supervised
- Supervised Fine-Tuning vs. RLHF: How to Choose the Right Approach (https://www.invisible.co/blog/supervised-fine-tuning-vs-rlhf-how-to right-approach-to-train-your-llm)
- Fine-Tuning vs RLHF: Choosing the Best LLM Training Method (https://cleverx.com/blog/supervised-fine-tuning-vs-rlhf-choosing-th to-train-your-llm)
- What is supervised fine-tuning? - BlueDot Impact (https://bluedot.org is-supervised-fine-tuning)

Process Reward Model (PRM) vs Outcome R Model (ORM)

Outcome Reward Model (ORM) = Trajectory Reward

- single reward at end of trajectory

Process Reward Model (PRM) = step by step reward

References for Process Reward Models vs Outcome Reward Models

- A Comprehensive Survey of Reward Models: Taxonomy and Applica (https://arxiv.org/html/2504.12328v1)
- Reward Modeling | RLHF Book by Nathan Lambert (https://rlhfboo reward-models.html)
- Let's Verify Step by Step (OpenAI, Process Supervision) (https://cdn.openai.com/improving-mathematical-reasoning-with-p supervision/Lets_Verify_Step_by_Step.pdf)
- Getting LLMs To Reason With Process Rewards (https://patmcguinness.substack.com/p/getting-llms-to-reason-wit

References

Title (linked)	Commentary
InstructGPT: Aligning Language Models with Human Intent via RLHF (https://arxiv.org/abs/2203.02155)	Foundational paper laying out the RLHF approach to align LLMs with human intent using human preference data. Essential for understanding RLHF theory and practice.
A Survey on Post-Training of Large Language Models (https://arxiv.org/abs/2503.06072)	Comprehensive survey reviewing SFT, RLHF, and newer alignment methods. Synthesizes research trends and challenges in LLM post-training.
Reinforcement Learning from AI Feedback (RLAIF): A Scalable Alternative to RLHF (https://arxiv.org/abs/2309.00267)	Introduces RLAIF, replacing human feedback with AI-generated feedback for scalable alignment. Critical for understanding automated feedback approaches.
Constitutional AI: Harmlessness from AI Feedback (https://www.anthropic.com/research/constitutional-ai-harmlessness-from-ai-feedback)	Proposes Constitutional AI, using a fixed ethical constitution for AI self-critique and revision to improve alignment without human labels.
LLM Post-Training: A Deep Dive into Reasoning Large Language Models (https://arxiv.org/abs/2502.21321)	Examines post-training methods focused on improving reasoning in LLMs via SFT and RL, analyzing mechanics and challenges.
SFT Memorizes, RL Generalizes: A Comparative Study of Post-Training Methods for LLMs (https://arxiv.org/abs/2501.17161)	Empirically compares SFT and RL in LLMs, showing SFT excels at memorization while RL generalizes better and improves alignment.
How Reinforcement Learning Beats Supervised Fine-Tuning When Data Is Scarce (https://predibase.com/blog/how-reinforcement-learning-beats-supervised-fine-tuning-when-data-is-scarce)	Blog explaining why RL methods can outperform SFT in low-data regimes; offers practical insights for training efficiency.
Beyond Next-Token Prediction: How Post-Training Teaches LLMs to Reason (https://toloka.ai/blog/how-post-training-teaches-llms-to-reason/)	Discusses how combining SFT and RL post-training enables complex reasoning in LLMs, with examples and experimental findings.
Demystifying Reasoning Models (https://cameronrwolfe.substack.com/p/demystifying-reasoning-models)	Blog unpacking the roles of SFT and RL in reasoning capability development; bridges theory and practice with clear explanations.
RLHF vs RLAIF: A Detailed Comparison of AI Training Methods (https://www.sapien.io/blog/rlaif-vs-rlhf-understanding-the-differences)	Detailed comparison of RLHF and RLAIF approaches, illustrating differences in feedback sources and workflows for AI alignment.

In [2]:
```
print("Done")
```

```
Done
```